# The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators

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HarnessRSI 70 ★★★

evidence: improver

Paper

## 1 What changes, what is kept

**One node = one shared, editable workspace**

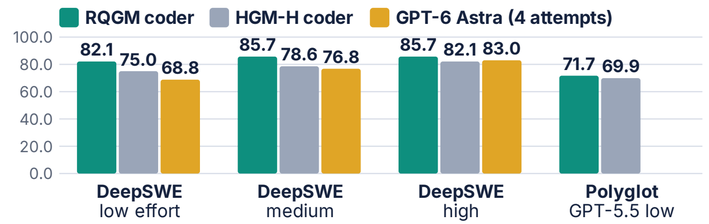
| | | | |
|---|---|---|---|
| Agents | task solvers | Evaluators | reviewer, grader |
| Meta-agent | edits all roles | Fixed | scoring harness |

1. **Expand + evaluate** — parents picked by clade metaproductivity; a frozen evaluator scores agents
2. **Checkpoint (powers of 2)** — challengers vs incumbent on a ground-truth anchor
3. **Replace only if better** — higher anchor best-belief wins (pool slots: anchor + pool-rejection score)
4. **Selective erasure** — drop records that depended on the displaced evaluator
5. **Adversarial pool (optional)** — outputs the old evaluator accepted must be rejected next epoch

**selection + replacement use the validation split only**
writer / prover have no anchor: chosen by the system's own evaluators, then scored post hoc on a test split unseen in search

Paper Fig. 9 says only anchor evidence gates replacement; its App. F.2.1 adds the pool term. DeepSWE meta-agent: high effort.

## 2 Headline: held-out pass rate (%)




28 / 166 held-out tasks; single runs ("eight" = 6 arms listed); ‡ assumes 1 attempt; DeepSWE meta-agent: high effort.

| Other domains (single runs) | HGM-H | RQGM |
|---|---|---|
| Paper writing: panel mean accept. % | 21.8 | **38.8 · 40.5** |
| Proof: panel mean /7 (IMO25 4.07) | 3.73 | **4.33** |

## 3 Controls that decide if the gain is real

**Ablation · reviewer anchor acc. % (Table 11)**




Nemotron 3 Super, paper domain; proof ablation also exists; none for coding. Full, no-pool and no-erasure intervals overlap.

| Search tokens to winner | RQGM | HGM-H |
|---|---|---|
| DeepSWE coder | **587M** | 429M |
| Paper writer: specialist | **221.8M** | 42.5M |
| Paper writer: generalist | 44.6M | 42.5M |
| Proof: specialist · generalist | **88.0M · 37.9M** | 21.5M |

Winners cost more tokens than HGM-H (DeepSWE 1.37‡×), yet HGM-H ran on to 2532.7M (RQGM 2876.8M) with no better coder. Generalists are near-matched (proof 1.76‡×).

| Proof score by grader | IMO25 | HGM-H | RQGM spec. |
|---|---|---|---|
| ProofAutoGrader | 4.05 | 3.55 | 3.85 |
| HGM-H grader | 4.10 | 3.65 | 4.20 |
| RQGM grader (co-evolved) | 4.05 | 4.00 | **4.95** |
| **Without RQGM grader** | 4.08 | 3.60 | 4.03 |

The RQGM grader co-evolved with the specialist and sits in the panel. DeepSWE reviewers before the last accept 81.8–100.0% of failing patches; the final one 20.8% (anchor rises are what the rule selects for).

## 4 Where the gain is absent or thin

- **DeepSWE high:** 85.7 vs 83.0 Astra, 82.1 HGM-H = 1 task‡
- **Polyglot:** 71.7 vs 69.9 = 3 of 166 tasks‡
- **Proof:** 4.33 ±0.41 vs 4.07 ±0.42; Pass@7 48.3% vs 55.0%
- **APReS held-out accuracy:** RQGM reviewers 80–84% vs 88%
- **DeepSWE winner** built under reviewers with almost no signal; HGM-H made the same regression-test edit
- Sakana reviewer: 1.0 / 2.0 / 5.0 %, inside ±2.2 to ±4.3
- Reward design: +0.059‡ vs static std 0.128

## Takeaway

Evolving reviewers and graders behind a labelled anchor can supply a search signal where no benchmark exists. Evidence on the evaluators is mixed (held-out APReS reviewers 80–84% vs HGM-H 88%); agent gains are single runs, a few coding tasks wide, and partly scored by the system's own evaluators.

## Our read · HarnessRSI

**Score 70 · ★★★ · evidence level: improver**

**Reports:** held-out splits, effort transfer, token cost, 2 ablations.
**Does not report:** repeated headline runs, held-out score at matched tokens, coding ablation, contamination test.